

and then replicate the configuration on a large-scale redundant deployment. We also need to monitor the host disk space very frequently to avoid storage failures and monitor VM instances regularly for compute host performance.

CloudStack: Containerisation

As per last year's *Forbes* report, container adoption is growing faster than expected. A *Gartner* report states that by 2023 there will be more applications that run in production with containers, and around 70 per cent of enterprises will run more than two container applications (at least) in production. Hence, there is a growing demand for containerised solutions across industries/topologies/geographies, and across CSPs and cloud adoption models (hybrid and multi-cloud).

StackRox has done extensive research on container adoption strategy and its trends. Six of these trends are shared below.

- Faster container adoption by organisations, which increased by 22 per cent last year. Organisations prefer the head-first approach to containerise applications as part of their digital transformation journey.
- The growth rate of containerised applications moving to the cloud was 32 per cent in the last six months. This indicates that enterprises prefer containers not only for building but also for testing applications.
- AWS container adoption is the most popular among customers across industry, followed by Azure and GCP. AWS leads with a 28-35 per cent share in container adoption.
- Container security, DevOps integration, open source container platforms and container management are the key focus in building a container adoption strategy.
- Security as Code and Infrastructure as Code are the key solution drivers in building a resilient container adoption blueprint to have a ready-to-deploy infra configuration.

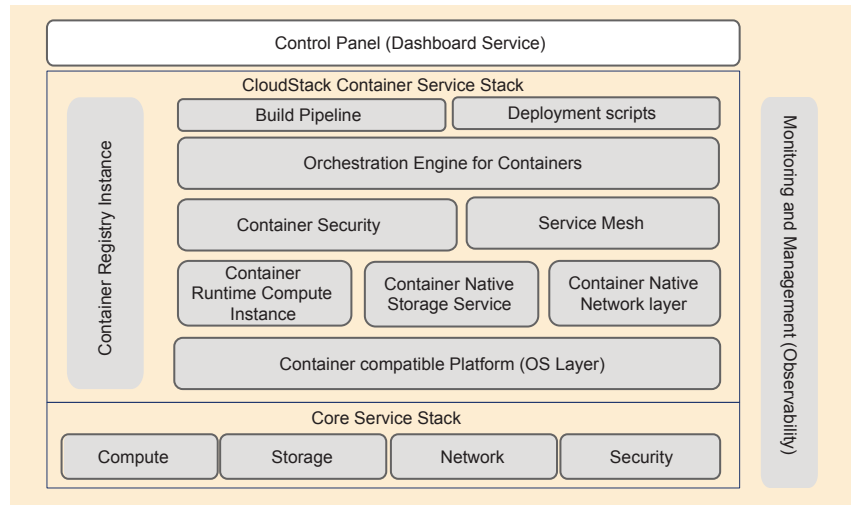


Figure 3: Multi-zone region deployment of CloudStack services

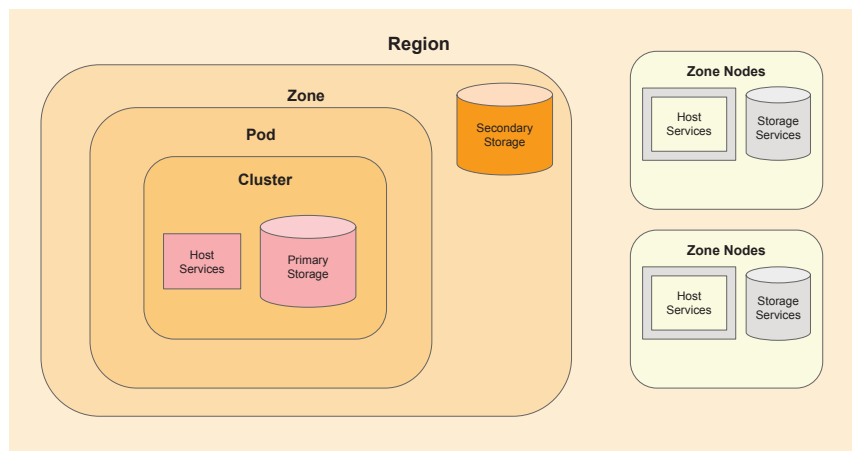


Figure 4: Component architecture of CloudStack Kubernetes Service

- Containers need to integrate with existing infrastructure and should not demand higher investment in infrastructure replacement for on-premise or cloud services. Apache CloudStack was formally developed as a cloud infrastructure management service for hybrid cloud solutions. It later adopted support for container deployment for hybrid cloud environments with the plug-in called CloudStack Kubernetes Service (CKS). CKS has a powerful user interface called the Control Panel or Dashboard service, using which one can build, deploy and monitor container instances quickly.

CKS is a multi-tenant solution, where multiple customers or entities share a common workload environment or platform with highly secured restrictions between their shared infrastructure components. They can build and deploy cloud-native container images and handle container cluster management using advanced cluster management tools.

CloudStack: Cloud service expense management

Cloud service expense management (CSEM) is a new-age requirement from cloud service platforms to provide cost transparency and cost management